CRITICAL CARE MEDICINE · MIDTERM · COMPLETE QUESTION BANK

Critical Care Medicine — Midterm All Lectures Question Bank

طب الحالات الحرجة — منهج الميدترم — مجمّع أسئلة المادة كلها

4	124	61	60
Lectures	Past exam questions	NCLEX-style	Flashcards

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المصادر:

أسئلة سابقة

كويزات سابقة (طب حالات حرجة)

ترقيم الأسئلة: رقم المحاضرة ثم رقم السؤال (مثلاً 4.12). أسئلة NCLEX عليها حرف N. الإجابات كلها في آخر الملف.

اللَّهُمَّ لا سَهْلَ إِلَّا ما جَعَلْتَهُ سَهْلًا، وأنتَ تَجْعَلُ الحَزْنَ إذا شِئْتَ سَهْلًا — صلِّ على النبي ﷺ

LECTURE 1

Acute Coronary Syndrome (ACS)

متلازمة الشريان التاجي الحادة

Part A · Past Exam Questions

أسئلة الدفعات السابقة

1.1 Acute Coronary Syndrome term includes STEMI and Non STEMI only. (كويزات سابقة (طب حالات حرجة)

a. True **b.** False

1.2 The definitive treatment for patient with ST-elevation Myocardial Infarction (STEMI) is ..... (كويزات سابقة (طب حالات حرجة)

a. Oxygen
b. Nitrates
c. Coronary angiography and coronary revascularization
d. Good hydration

1.3 Treatment of Acute Coronary Syndrome doesn't include: (كويزات سابقة (طب حالات حرجة)

a. Antiplatelet drugs **b.** Nitrates
c. Beta blocker drugs **d.** Aminophylline

1.4 Differentiation between Myocardial Infarction and Angina could be done through: (كويزات سابقة (طب حالات حرجة)

a. Chest pain characteristics **b.** Checking vital signs
c. Checking cardiac enzymes **d.** Identification of risk factors

1.5 The definitive treatment for patient with ST elevation Myocardial Infarction (STEMI) is: (كويزات سابقة (طب حالات حرجة)

a. Coronary angiography & revascularization
b. Good hydration
c. Oxygen
d. Nitrates

1.6 Management of acute ST elevation MI includes morphine, oxygen, nitrates only. (كويزات سابقة (طب حالات حرجة)

a. True **b.** False

1.7 GRACE ACS risk includes all of the following except: (كويزات سابقة (طب حالات حرجة)

a. Age **b.** Heart rate
c. Sodium concentration **d.** Systolic blood pressure

1.8 **Differentiation between Myocardial Infarction and Angina could be done through checking:** كويزات سابقة (طب حالات حرجة)

a. Vital signs
b. Pain characteristics
c. Cardiac enzymes
d. Risk factors

1.9 **What does Acute Coronary Syndrome (ACS) encompass?** أسئلة سابقة

a. Only unstable angina
b. A spectrum of conditions including unstable angina, NSTEMI, and STEMI
c. Only ST-Elevation Myocardial Infarction (STEMI)
d. A condition only seen in older adults

1.10 **Which of the following is the hallmark symptom of Acute Coronary Syndrome?** أسئلة سابقة

a. Abdominal pain
b. Chest pain, often described as pressure, tightness, or heaviness
c. Dizziness
d. Hemorrhage

1.11 **What laboratory test is most useful in diagnosing ACS?** أسئلة سابقة

a. Complete Blood Count (CBC)
b. Cardiac Enzymes (e.g., Troponin)
c. Liver Function Tests
d. Arterial Blood Gas (ABG)

1.12 **What ECG change is commonly seen in Unstable Angina (UA) and NSTEMI?** أسئلة سابقة

a. ST segment elevation
b. T-wave inversion or ST segment depression
c. Q waves
d. No changes

1.13 **What is the main difference between NSTEMI and STEMI?** أسئلة سابقة

a. NSTEMI involves complete occlusion of a coronary artery, whereas STEMI does not
b. NSTEMI involves partial occlusion of a coronary artery, whereas STEMI involves complete occlusion
c. STEMI is less severe than NSTEMI
d. STEMI does not require any treatment

1.14 **Which of the following is a typical trigger for chest pain in ACS?** أسئلة سابقة

a. Physical activity, emotional stress, or cold weather
b. Sleeping
c. Eating meals
d. High temperature

1.15 **In ACS, which of the following is used to relieve chest pain?** أسئلة سابقة

a. Exercise
b. Nitroglycerin
c. Corticosteroids
d. Antihistamines

1.16 What does the TIMI (Thrombolysis in Myocardial Infarction) Risk Score help determine? أسئلة سابقة

a. The presence of ST-segment elevation
b. The prognosis of death and ischemic events in ACS patients
c. The exact location of plaque rupture
d. The number of coronary arteries involved

1.17 What does the GRACE Risk Model in ACS assess? أسئلة سابقة

a. The likelihood of developing hypertension
b. The risk of in-hospital mortality
c. The severity of atherosclerotic plaques
d. The occurrence of arrhythmias

1.18 Which of the following is a common symptom of STEMI? أسئلة سابقة

a. Mild chest discomfort
b. Severe chest pain with radiation, nausea, and sweating
c. Shortness of breath only
d. Pain that is relieved by rest

1.19 Which of the following ECG changes is commonly seen in STEMI? أسئلة سابقة

a. Significant ST-segment elevation in ≥2 contiguous leads
b. Q waves without ST-segment changes
c. T-wave flattening only
d. ST-segment depression only

1.20 What does elevated troponin levels indicate in the context of ACS? أسئلة سابقة

a. Dehydration
b. Myocardial necrosis
c. Hyperglycemia
d. Kidney failure

1.21 What is the key feature distinguishing Unstable Angina (UA) from Stable Angina? أسئلة سابقة

a. The pain occurs at rest or with minimal exertion in UA
b. Stable angina is relieved by nitroglycerin, while UA is not
c. UA is not associated with pain
d. Stable angina does not cause chest pain

1.22 Which treatment is part of the initial management of ACS? أسئلة سابقة

a. Beta-blockers only
b. MONA (Morphine, Oxygen, Nitroglycerin, Antithrombotic therapy)
c. Corticosteroids
d. Sodium bicarbonate

1.23 What is the purpose of nitroglycerin in ACS management? أسئلة سابقة

a. To increase blood pressure
b. To reduce myocardial oxygen demand
c. To prevent arrhythmias
d. To dissolve blood clots

1.24 Which of the following is an important part of interventional management for ACS? أسئلة سابقة

a. Coronary Angiography followed by PCI (Percutaneous Coronary Intervention)
b. Long-term bed rest
c. Administration of corticosteroids
d. Blood transfusion

1.25 Which surgical intervention is commonly used for severe cases of coronary artery disease? أسئلة سابقة

a. Coronary Artery Bypass Grafting (CABG)
b. Stent placement only
c. Percutaneous balloon angioplasty
d. Valve replacement surgery

1.26 Which of the following mechanical assist devices is used for hemodynamic support in severe cases of ACS? أسئلة سابقة

a. V-A ECMO
b. Ventricular assist device (VAD)
c. Intra-aortic balloon pump (IABP)
d. Continuous positive airway pressure (CPAP)

1.27 Which device is considered less invasive than ECMO and used for circulatory support during PCI? أسئلة سابقة

a. Impella
b. IABP
c. V-A ECMO
d. Defibrillator

1.28 What does the Intra-Aortic Balloon Pump (IABP) do? أسئلة سابقة

a. Supports both heart and lung function
b. Improves blood flow to the heart and reduces workload during PCI
c. Treats respiratory failure
d. Provides long-term mechanical ventilation

1.29 Which of the following is a typical ECG finding in NSTEMI? أسئلة سابقة

a. Significant ST-segment elevation
b. No change in ST-segment
c. ST-segment depression or T-wave inversion
d. Q waves

1.30 What role does the Killip Class play in ACS? أسئلة سابقة

a. It is a classification of heart failure severity, which impacts mortality risk
b. It indicates the type of coronary artery blocked
c. It determines the number of PCI required
d. It is used to assess the extent of myocardial injury

1.31 What does the GRACE Risk Score assess in ACS patients? أسئلة سابقة

a. Risk of arrhythmias only
b. Risk of in-hospital mortality
c. Length of hospital stay
d. Response to antithrombotic therapy

1.32 What is the most appropriate treatment for STEMI when PCI is unavailable? أسئلة سابقة

a. Immediate coronary artery bypass grafting (CABG)
b. Thrombolytics
c. Long-term bed rest
d. Corticosteroids

1.33 What is the primary cause of myocardial ischemia in ACS? أسئلة سابقة

a. Hypoxia
b. Atherosclerotic plaque rupture and thrombus formation
c. Viral infection
d. High blood pressure

1.34 How does morphine assist in ACS management? أسئلة سابقة

a. It increases heart rate
b. It provides pain and anxiety relief
c. It prevents clot formation
d. It increases blood pressure

1.35 In the context of ACS, what is the primary action of aspirin? أسئلة سابقة

a. Increase platelet aggregation
b. Prevent thrombus formation by inhibiting platelet aggregation
c. Dilate blood vessels
d. Reduce blood pressure

1.36 What is a characteristic finding in the cardiac biomarkers of NSTEMI? أسئلة سابقة

a. Normal troponin levels
b. Elevated troponin levels
c. Normal creatine kinase levels
d. Decreased B-type natriuretic peptide levels

1.37 What does partial coronary artery occlusion typically result in during NSTEMI? أسئلة سابقة

a. Full-thickness myocardial infarction
b. Transient reversible ischemia
c. Total coronary artery blockage
d. No myocardial injury

1.38 Which of the following is a common sign of cardiogenic shock in ACS? أسئلة سابقة

a. High blood pressure
b. Low blood pressure with signs of organ dysfunction
c. Normal heart rate and rhythm
d. Lack of chest pain

1.39 What is the next step in management if a 12-lead ECG shows STEMI after ROSC? أسئلة سابقة

a. Administer amiodarone
b. Perform targeted temperature management
c. Proceed with coronary angiography and revascularization
d. Start advanced airway management

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

1.N1 A patient has chest pain at rest, ST depression on ECG, and elevated troponin. The most likely diagnosis is: NCLEX

a. Stable angina
b. Unstable angina
c. NSTEMI
d. STEMI

1.N2 ACS includes (select all): SELECT ALL

a. Unstable angina
b. NSTEMI
c. STEMI
d. Stable angina
e. Pericarditis

1.N3 The cardiac biomarker most specific for myocardial injury is: NCLEX

a. CK-total
b. Troponin I/T
c. LDH
d. Myoglobin

1.N4 A patient with STEMI arrives at a PCI-capable hospital. The target door-to-balloon time is: NCLEX

a. ≤ 90 minutes
b. ≤ 6 hours
c. ≤ 24 hours
d. ≤ 12 hours

1.N5 Before giving nitroglycerin to a patient with ACS, the nurse must check (select all): SELECT ALL

a. Systolic BP > 90 mmHg
b. No recent phosphodiesterase inhibitor (sildenafil)
c. No right ventricular infarction
d. Serum potassium
e. Blood glucose

1.N6 Which drug inhibits platelet aggregation and should be chewed early in suspected ACS? NCLEX

a. Aspirin 162–325 mg
b. Paracetamol
c. Morphine
d. Furosemide

1.N7 Contraindications to thrombolytic therapy include (select all): SELECT ALL

a. Previous hemorrhagic stroke
b. Active internal bleeding
c. Suspected aortic dissection
d. Age above 50 years
e. Recent major surgery or trauma

1.N8 Beta-blockers in ACS work mainly by: NCLEX

a. Increasing heart rate
b. Reducing myocardial oxygen demand
c. Dissolving the clot
d. Raising blood pressure

1.N9 Killip class IV in a patient with MI means: NCLEX

a. No heart failure
b. Basal crackles / S3
c. Frank pulmonary edema
d. Cardiogenic shock

1.N10 **Signs of reperfusion after thrombolysis include (select all):** SELECT ALL

a. Relief of chest pain
b. Resolution of ST elevation
c. Reperfusion arrhythmias
d. Rising ST segment
e. Increasing chest pain

1.N11 **Which nursing action has priority during the first hour for a patient with STEMI?** NCLEX

a. Teach about diet
b. Continuous ECG monitoring with IV access and pain relief
c. Ambulate the patient
d. Schedule an exercise test

1.N12 **Women, older adults and diabetics with ACS may present with:** NCLEX

a. Atypical symptoms such as dyspnea, fatigue or epigastric pain
b. Always crushing chest pain
c. Fever and rash
d. Bradycardia only

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

ACS	Spectrum due to acute myocardial ischemia: unstable angina, NSTEMI, STEMI.	**Pathophysiology**	Atherosclerotic plaque rupture → platelet aggregation → thrombus → partial or total occlusion.
Unstable angina	Chest pain at rest / new / worsening; ECG may show ST depression; troponin normal.	**NSTEMI**	Partial occlusion; ST depression or T inversion; troponin raised.
STEMI	Complete occlusion; ST elevation ≥ 2 contiguous leads (or new LBBB); troponin raised.	**Typical pain**	Retrosternal pressure > 20 min, radiating to jaw/left arm, with sweating, nausea, dyspnea; not relieved by rest.
Troponin	Rises 3–4 h, peaks ~24 h, stays raised 7–14 days — most specific marker.	**MONA + BAH**	Morphine, Oxygen (if SpO_2 < 90%), Nitrates, Aspirin + Beta-blocker, Anticoagulant (heparin), High-intensity statin.
Definitive STEMI treatment	Coronary angiography + PCI (door-to-balloon ≤ 90 min); thrombolysis if PCI unavailable (door-to-needle ≤ 30 min).	**CABG**	Surgical bypass for left main or multivessel disease.
TIMI score	Predicts risk of death / ischemic events.	**GRACE score**	Predicts in-hospital mortality: age, HR, SBP, creatinine, Killip class, cardiac arrest, ST deviation, enzymes (no sodium).

Killip class	I none · II crackles/S3 · III pulmonary edema · IV cardiogenic shock.
Not used in ACS	Aminophylline, corticosteroids.
Mechanical support	IABP, Impella, V-A ECMO for cardiogenic shock.

LECTURE 2

ECG Monitoring

مراقبة رسم القلب

Part A · Past Exam Questions

أسئلة الدفعات السابقة

2.1 **Ventricular Tachycardia is a shockable rhythm.** كويزات سابقة (طب حالات حرجة)

a. True **b.** False

2.2 **Electrocardiography (ECG) is considered as a diagnostic tool for STEMI.** كويزات سابقة (طب حالات حرجة)

a. True **b.** False

2.3 **The following rhythm is considered shockable rhythm during cardiac arrest:** كويزات سابقة (طب حالات حرجة)

a. Asystole
b. Ventricular Fibrillation
c. Pulseless electrical activity (PEA)
d. Premature Ventricular Contraction (PVC)

2.4 **In advanced cardiac life support, shockable rhythms include:** كويزات سابقة (طب حالات حرجة)

a. Ventricular tachycardia in hemodynamically stable patient
b. Pulseless ventricular tachycardia
c. Premature ventricular tachycardia
d. B and C

2.5 **Pulseless ventricular tachycardia is considered from non-shockable rhythms during advanced cardiac life support.** كويزات سابقة (طب حالات حرجة)

a. True **b.** False

2.6 **What is a typical finding on an ECG in a patient with PE?** أسئلة سابقة

a. Left bundle branch block **b.** Sinus tachycardia
c. Normal sinus rhythm **d.** ST-segment elevation

2.7 **Which of the following investigations can identify arrhythmias that could lead to embolic stroke?** أسئلة سابقة

a. CT scan **b.** Duplex Ultrasound
c. Electrocardiogram (ECG) **d.** MRI

2.8 **Which rhythm is considered shockable in ACLS?** أسئلة سابقة

a. Asystole **b.** Pulseless Electrical Activity (PEA)
c. Ventricular Fibrillation (VF) **d.** Normal sinus rhythm

2.9 What is the key difference between shockable and non-shockable rhythms in ACLS? أسئلة سابقة

a. Shockable rhythms require defibrillation, non-shockable rhythms do not
b. Shockable rhythms are always treated with epinephrine
c. Non-shockable rhythms respond to defibrillation
d. Shockable rhythms require higher doses of epinephrine

2.10 What is the first action to take if a patient shows signs of a shockable rhythm like VF or VT? أسئلة سابقة

a. Administer epinephrine
b. Perform CPR immediately
c. Perform defibrillation
d. Insert an advanced airway

2.11 When is defibrillation indicated in a patient with a non-shockable rhythm like PEA or Asystole? أسئلة سابقة

a. Immediately after CPR
b. After 1 minute of CPR
c. Never
d. Only after the administration of epinephrine

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

2.N1 On standard ECG paper, one small square equals: NCLEX

a. 0.04 seconds
b. 0.2 seconds
c. 1 second
d. 0.12 seconds

2.N2 The P wave on the ECG represents: NCLEX

a. Ventricular depolarization
b. Atrial depolarization
c. Ventricular repolarization
d. Atrial repolarization

2.N3 The normal PR interval is: NCLEX

a. 0.04–0.08 s
b. 0.12–0.20 s
c. 0.24–0.40 s
d. > 0.20 s

2.N4 The normal QRS duration is: NCLEX

a. < 0.12 seconds
b. 0.12–0.20 seconds
c. > 0.20 seconds
d. Exactly 0.40 seconds

2.N5 Using the 1500 method, if there are 15 small squares between two R waves, the heart rate is: NCLEX

a. 60 bpm
b. 100 bpm
c. 150 bpm
d. 75 bpm

2.N6 For an irregular rhythm, the best way to calculate heart rate is: NCLEX

a. 1500 method
b. 300 method
c. Count QRS complexes in a 6-second strip × 10
d. Measure the PR interval

2.N7 Correct placement of the V1 chest lead is: NCLEX

a. 4th intercostal space, right sternal border
b. 5th intercostal space, midclavicular line
c. 4th intercostal space, left sternal border
d. Left anterior axillary line

2.N8 Steps of systematic rhythm interpretation include (select all): SELECT ALL

a. Determine the rate
b. Assess regularity
c. Examine P waves
d. Measure PR and QRS
e. Check the patient's blood sugar

2.N9 Sinus bradycardia is defined as a sinus rhythm with a rate: NCLEX

a. < 60 bpm
b. < 80 bpm
c. > 100 bpm
d. < 40 bpm only

2.N10 An ECG shows an irregularly irregular rhythm with no P waves and fibrillatory baseline. This is: NCLEX

a. Atrial flutter
b. Atrial fibrillation
c. Sinus arrhythmia
d. Third-degree block

2.N11 A 'saw-tooth' pattern of flutter waves is characteristic of: NCLEX

a. Atrial flutter
b. Ventricular fibrillation
c. Asystole
d. First-degree AV block

2.N12 Wide, bizarre, early QRS complexes without a preceding P wave are: NCLEX

a. Premature atrial contractions
b. Premature ventricular contractions (PVCs)
c. Junctional rhythm
d. Sinus beats

2.N13 The monitor shows ventricular tachycardia. The nurse's FIRST action is: NCLEX

a. Defibrillate immediately
b. Check the patient's pulse and responsiveness
c. Give amiodarone
d. Record a 12-lead ECG

2.N14 **Chaotic, irregular waveform with no identifiable P, QRS or T waves is:** NCLEX

a. Ventricular fibrillation
b. Atrial fibrillation
c. Sinus tachycardia
d. Junctional rhythm

2.N15 **Before defibrillating, the nurse should first:** NCLEX

a. Confirm the flat line in two leads
b. Check the patient and make sure everyone is clear
c. Give atropine
d. Synchronize the shock

2.N16 **A flat line appears on the monitor. The nurse should first:** NCLEX

a. Defibrillate
b. Check the patient, leads and gain; confirm in another lead
c. Give amiodarone
d. Synchronize cardioversion

2.N17 **Pulseless electrical activity (PEA) means:** NCLEX

a. Organized electrical activity on the monitor without a palpable pulse
b. Flat line
c. VF with a pulse
d. Normal rhythm with hypotension

2.N18 **Which rhythms are shockable? (select all)** SELECT ALL

a. Ventricular fibrillation
b. Pulseless ventricular tachycardia
c. Asystole
d. PEA
e. Sinus bradycardia

2.N19 **Third-degree (complete) heart block is recognized by:** NCLEX

a. P waves and QRS complexes beating independently (AV dissociation)
b. Progressive PR prolongation then a dropped beat
c. Constant PR > 0.20 s
d. Irregularly irregular rhythm

2.N20 **Progressive lengthening of the PR interval until a QRS is dropped describes:** NCLEX

a. Mobitz I (Wenckebach) second-degree block
b. Mobitz II block
c. First-degree block
d. Atrial flutter

2.N21 **Peaked (tall tented) T waves on the monitor suggest:** NCLEX

a. Hypokalemia
b. Hyperkalemia
c. Hypocalcemia
d. Hypothermia

2.N22 ECG evidence of STEMI is: NCLEX

a. ST elevation in ≥ 2 contiguous leads or new LBBB
b. ST depression in one lead
c. Tall P waves
d. Prolonged PR interval

2.N23 Nursing responsibilities in continuous cardiac monitoring include (select all): SELECT ALL

a. Clean and dry skin before electrode placement
b. Change electrodes every 24 hours or as needed
c. Set alarm limits and never switch them off
d. Check the patient, not just the monitor
e. Silence alarms that keep ringing

2.N24 Synchronized cardioversion differs from defibrillation because it: NCLEX

a. Delivers the shock on the R wave
b. Is used for VF
c. Uses no energy
d. Is given without consent in stable patients

2.N25 Sinus tachycardia is a regular sinus rhythm with a rate of: NCLEX

a. > 100 bpm
b. 60–100 bpm
c. < 60 bpm
d. > 300 bpm

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

ECG paper	Small square 0.04 s / 1 mm; large square 0.2 s / 5 mm; 10 mm = 1 mV.	**Waves**	P = atrial depolarization · QRS = ventricular depolarization · T = ventricular repolarization.
Normal intervals	PR 0.12–0.20 s · QRS < 0.12 s · QTc < 0.44 s.	**Heart rate**	300 ÷ large squares, 1500 ÷ small squares, or QRS in 6 s × 10 (irregular).
5-step interpretation	Rate → rhythm (regularity) → P waves → PR interval → QRS width.	**Limb leads**	RA white, LA black, LL red, RL green ('white on right, smoke over fire').
Chest leads	V1 4th ICS right sternal border · V2 4th ICS left · V4 5th ICS MCL · V6 mid-axillary.	**Sinus brady / tachy**	< 60 / > 100 bpm with normal P before each QRS.
Atrial fibrillation	Irregularly irregular, no P waves; stroke risk → anticoagulation.	**Atrial flutter**	Saw-tooth F waves, atrial rate ~300.

PVC	Early wide bizarre QRS without P; dangerous if frequent, multifocal, R-on-T.
VT	≥ 3 wide QRS, rate > 100; pulseless → shock; with pulse unstable → synchronized cardioversion.
VF	Chaotic baseline, no output → defibrillate + CPR.
Asystole / PEA	Non-shockable → CPR + epinephrine 1 mg every 3–5 min; search Hs & Ts.
Heart blocks	1st: PR > 0.20 · Mobitz I: PR lengthens then dropped beat · Mobitz II: constant PR, dropped beats · 3rd: AV dissociation → pacing.
STEMI on ECG	ST elevation in ≥ 2 contiguous leads; II, III, aVF inferior · V1–V4 anterior · I, aVL, V5–V6 lateral.
Potassium	Hyperkalemia: peaked T, wide QRS · Hypokalemia: flat T, U waves.

LECTURE 3

Mechanical Ventilation

التنفس الصناعي

Part A · Past Exam Questions

أسئلة الدفعات السابقة

3.1 Mechanical ventilation is best described as: (كويزات سابقة (طب حالات حرجة)

a. Invasive or non-invasive
b. A permanent device
c. A new technology
d. Doesn't induce lung injury

3.2 Intrinsic PEEP, oxygen toxicity and pneumothorax are complications of mechanical ventilation. (كويزات سابقة (طب حالات حرجة)

a. True
b. False

3.3 A protective lung strategy in mechanical ventilation is to set TV at ..... to ..... ml/kg. (كويزات سابقة (طب حالات حرجة)

a. 1-3
b. 4-6
c. 7-10
d. 11-13

3.4 Which of the following are assist modes of ventilation? 1. Volume control 2. Pressure control 3. SIMV 4. CPAP (كويزات سابقة (طب حالات حرجة)

a. 1,3,4
b. 1,2,4
c. 2,3,4
d. 1,2,3

3.5 What is the recommended angle for the head of the bed in ventilated coma patients? أسئلة سابقة

a. 0 degrees
b. 15 to 30 degrees
c. 30 to 45 degrees
d. 60 to 90 degrees

3.6 What is the purpose of a "Ventilator Bundle" in coma management? أسئلة سابقة

a. To prevent skin ulcers
b. To reduce the need for sedation
c. To prevent respiratory and circulatory complications
d. To ensure proper hydration

3.7 What does "ventilator bundle" refer to in the context of coma management? أسئلة سابقة

a. Measures for preventing respiratory and circulatory complications in ventilated patients
b. A treatment for electrolyte imbalance
c. A medication plan for coma patients
d. A nutritional intervention for coma patients

3.8 What is the primary goal of non-invasive ventilation in COPD patients during acute exacerbations? أسئلة سابقة

a. To prevent oxygenation failure
b. To improve blood sugar levels
c. To decrease sputum production
d. To increase lung volume

3.9 What is the goal of using non-invasive ventilation (NIV) in ARDS? أسئلة سابقة

a. To decrease blood pressure
b. To support lung function without intubation
c. To provide sedatives
d. To treat pneumonia

3.10 What is the most important component of ventilator care in ARDS patients? أسئلة سابقة

a. Use of high oxygen levels
b. Low tidal volumes and proper PEEP
c. Administration of antibiotics
d. Use of steroids for inflammation

3.11 What should be monitored closely in patients with ARDS who are on mechanical ventilation? أسئلة سابقة

a. Platelet count
b. Oxygenation and ventilation parameters
c. Blood glucose levels
d. Renal function only

3.12 What is the primary purpose of mechanical ventilation (MV)? أسئلة سابقة

a. To directly treat illnesses
b. To help a patient breathe when they cannot breathe on their own
c. To decrease blood pressure
d. To eliminate toxins from the blood

3.13 Which of the following is a non-invasive form of mechanical ventilation? أسئلة سابقة

a. Intubation
b. High Flow Nasal Cannula (HFNC)
c. Tracheostomy
d. Invasive Positive Pressure Ventilation (IPPV)

3.14 Which condition is an indication for mechanical ventilation? أسئلة سابقة

a. Mild Asthma exacerbation
b. Insufficient oxygenation (hypoxemia)
c. Controlled diabetes
d. Minor allergic reactions

3.15 What is the primary benefit of mechanical ventilation in cases of insufficient ventilation? أسئلة سابقة

a. It helps in removing excess oxygen
b. It aids in the removal of carbon dioxide (CO_2)
c. It helps reduce airway inflammation
d. It increases the amount of oxygen absorbed by the lungs

أسئلة سابقة

3.16 **In which of the following situations would mechanical ventilation be needed for airway protection?**

a. Mild respiratory infection
b. Patient at risk of aspiration
c. Stable asthmatic patient
d. After a minor trauma

أسئلة سابقة

3.17 **Barotrauma is a complication of mechanical ventilation. What causes it?**

a. Injury from excessive tidal volume and airway pressure
b. High oxygen levels causing tissue damage
c. Infection from the ventilator
d. Excessive CO_2 buildup in the lungs

أسئلة سابقة

3.18 **Auto-PEEP is a phenomenon in mechanical ventilation. What causes it?**

a. Incomplete exhalation leading to positive pressure build-up
b. Excessive oxygen delivery during ventilation
c. Failure to initiate the first breath
d. Sudden reduction in ventilation rate

أسئلة سابقة

3.19 **Which of the following is not a type of mechanical ventilation?**

a. High Flow Nasal Cannula (HFNC)
b. Helmet
c. Intubation
d. External Ventilation Mask

أسئلة سابقة

3.20 **What is the most common indication for mechanical ventilation in severe asthma cases?**

a. Mild airway narrowing
b. Respiratory failure due to airway constriction
c. Chronic asthma management
d. Normal blood oxygen saturation levels

أسئلة سابقة

3.21 **What is the main purpose of using the Assist/Control (A/C) mode of ventilation?**

a. To allow the patient to breathe spontaneously
b. To provide full ventilator support if the patient cannot breathe on their own
c. To help with the timing of spontaneous breaths
d. To assist the patient in weaning off mechanical support

أسئلة سابقة

3.22 **In which mode of mechanical ventilation does the ventilator provide mandatory breaths at set intervals while allowing spontaneous breathing?**

a. Assist/Control (A/C)
b. Synchronous Intermittent Mandatory Ventilation (SIMV)
c. Pressure Support Ventilation (PSV)
d. Continuous Positive Airway Pressure (CPAP)

أسئلة سابقة

3.23 **Pressure Support Ventilation (PSV) is most appropriate for which type of patient?**

a. A patient who is unable to breathe on their own
b. A patient who requires full ventilator support
c. A patient who can breathe independently but needs assistance with the effort
d. A patient with acute lung injury

أسئلة سابقة

3.24 **Which of the following is true regarding Continuous Positive Airway Pressure (CPAP) mode?**

a. It provides complete control of breathing
b. It keeps the airways open without controlling the patient's breaths
c. It allows for spontaneous breathing only
d. It is used only during sleep

أسئلة سابقة

3.25 **What is the main difference between Control Mode Ventilation (CMV) and other ventilation modes?**

a. The patient controls the timing of breaths
b. The ventilator provides all breaths with no spontaneous breathing allowed
c. The patient does not require any mechanical support
d. The mode alternates between high and low airway pressures

أسئلة سابقة

3.26 **Which of the following is not included in the Ventilator Bundle for reducing complications like VAP?**

a. Head of bed elevation
b. Oral care with chlorhexidine
c. Daily sedation vacation
d. Increased oxygen saturation

أسئلة سابقة

3.27 **What is the purpose of peptic ulcer prophylaxis in mechanically ventilated patients?**

a. To prevent respiratory infections
b. To prevent the development of stress ulcers
c. To promote lung expansion
d. To reduce inflammation in the lungs

أسئلة سابقة

3.28 **Which of the following complications is most commonly associated with mechanical ventilation for extended periods?**

a. Barotrauma
b. Pneumothorax
c. Ventilator-Associated Pneumonia (VAP)
d. All of the above

أسئلة سابقة

3.29 **High Flow Nasal Cannula (HFNC) provides what type of support in mechanical ventilation?**

a. High levels of oxygen delivery
b. Positive pressure ventilation
c. Complete mechanical ventilation support
d. Just oxygen delivery without pressure

أسئلة سابقة

3.30 **In which situation is intubation the most appropriate intervention?**

a. A patient requiring short-term oxygen support
b. A patient with moderate asthma exacerbation
c. A patient with severe airway obstruction or respiratory failure
d. A patient needing oxygen at home

أسئلة سابقة

3.31 **Which of the following is a benefit of mechanical ventilation?**

a. Decreases blood pressure
b. Increases patient activity levels
c. Facilitates carbon dioxide removal
d. Increases the production of red blood cells

أسئلة سابقة

3.32 **Ventilator-Induced Lung Injury (VILI) can occur due to which of the following?**

a. High PEEP and excessive tidal volume
b. Low PEEP and low tidal volume
c. Low-frequency ventilation
d. Short-term oxygen therapy

أسئلة سابقة

3.33 **The Synchronous Intermittent Mandatory Ventilation (SIMV) mode is most appropriate for patients who:**

a. Cannot breathe independently
b. Require full support for every breath
c. Are starting to breathe independently but need some ventilator assistance
d. Have complete lung collapse

أسئلة سابقة

3.34 **In which of the following conditions would Mechanical Ventilation be contraindicated?**

a. Severe asthma exacerbation
b. Acute lung injury (ALI)
c. Cardiac arrest with no chance of recovery
d. Acute respiratory distress syndrome (ARDS)

أسئلة سابقة

3.35 **Which ventilator mode is best suited for a patient who needs full ventilator support and cannot breathe on their own?**

a. Assist/Control (A/C)
b. Pressure Support Ventilation (PSV)
c. Synchronous Intermittent Mandatory Ventilation (SIMV)
d. Continuous Positive Airway Pressure (CPAP)

أسئلة سابقة

3.36 **Which complication of mechanical ventilation results from prolonged exposure to high oxygen levels?**

a. Pneumothorax
b. Oxygen toxicity
c. Barotrauma
d. Hypercapnia

3.37 What is the goal of Daily Sedation Vacation in ventilated patients? أسئلة سابقة

a. To increase the level of sedation
b. To assess whether sedation can be reduced or stopped
c. To adjust the ventilator settings
d. To assess the patient's need for mechanical ventilation

3.38 High PEEP levels during mechanical ventilation may lead to which complication? أسئلة سابقة

a. Oxygen toxicity
b. Pneumothorax
c. Oxygen desaturation
d. Hypercapnia

3.39 Which of the following ventilator alarms alerts the healthcare team to a potential issue with the patient's ventilation rate? أسئلة سابقة

a. High Pressure alarm
b. High Frequency alarm
c. Low Pressure alarm
d. Apnea alarm

3.40 Mechanical Ventilation can be required for patients with severe hypotension due to which condition? أسئلة سابقة

a. Sepsis
b. Heart failure
c. Cardiogenic shock
d. All of the above

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

3.N1 Indications for mechanical ventilation include (select all): SELECT ALL

a. Hypoxemic respiratory failure
b. Hypercapnic respiratory failure
c. Airway protection (GCS ≤ 8)
d. Cardiac arrest
e. Mild anxiety

3.N2 The high-pressure alarm on a ventilator may be caused by (select all): SELECT ALL

a. Secretions in the airway
b. Patient biting the tube or coughing
c. Kinked tubing
d. Disconnection of the circuit
e. Bronchospasm

3.N3 A ventilated patient suddenly desaturates. Using DOPE, the nurse checks for: NCLEX

a. Displacement, Obstruction, Pneumothorax, Equipment failure
b. Dehydration, Obesity, Pain, Edema
c. Drugs, Oxygen, Pressure, Electrolytes
d. Diet, Output, Position, Exercise

3.N4 Elements of the VAP prevention bundle include (select all): SELECT ALL

a. Head of bed 30–45°
b. Daily sedation interruption and readiness-to-wean assessment
c. Oral care with chlorhexidine
d. DVT and peptic ulcer prophylaxis
e. Routine daily circuit changes

3.N5 Endotracheal cuff pressure should be maintained at: NCLEX

a. 20–30 cmH_2O
b. 5–10 cmH_2O
c. 40–50 cmH_2O
d. 60 cmH_2O

3.N6 During suctioning of an intubated patient, the nurse should (select all): SELECT ALL

a. Pre-oxygenate with 100% O_2
b. Apply suction only while withdrawing
c. Limit each pass to 10–15 seconds
d. Use clean technique and reuse the catheter
e. Monitor SpO_2 and heart rate

3.N7 PEEP improves oxygenation by: NCLEX

a. Keeping alveoli open at end-expiration
b. Increasing respiratory rate
c. Reducing FiO_2 to zero
d. Increasing dead space

3.N8 Which ventilator mode delivers a set tidal volume for every breath, whether triggered by the patient or the machine? NCLEX

a. Assist/Control (A/C)
b. CPAP
c. Pressure support
d. SIMV

3.N9 Readiness for weaning is suggested by (select all): SELECT ALL

a. Underlying cause improving
b. Hemodynamic stability
c. $FiO_2 \leq 40–50\%$ with PEEP $\leq 5–8$
d. Adequate cough and mental status
e. High-dose vasopressors

3.N10 Signs of failure during a spontaneous breathing trial include (select all): SELECT ALL

a. RR > 35/min
b. $SpO_2 < 90\%$
c. Heart rate rising > 20% or arrhythmia
d. Diaphoresis and agitation
e. Calm, comfortable breathing

3.N11 Correct ETT position is confirmed by (select all): SELECT ALL

a. Waveform capnography ($ETCO_2$)
b. Bilateral equal breath sounds and chest rise
c. Chest X-ray
d. Patient talking normally
e. Absence of gastric sounds

3.N12 Lung-protective ventilation in ARDS uses tidal volume of: NCLEX

a. 4–6 (up to 8) mL/kg predicted body weight
b. 10–12 mL/kg actual weight
c. 15 mL/kg
d. 2 mL/kg

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

Purpose of MV	Support oxygenation and ventilation (CO_2 removal), protect the airway, rest respiratory muscles.
Indications	Hypoxemia, hypercapnia/acidosis, apnea, GCS ≤ 8 / aspiration risk, shock, cardiac arrest.
SIMV	Mandatory breaths synchronized with spontaneous breaths — used in weaning.
CPAP	Continuous pressure, all breaths spontaneous; keeps airways open.
Protective ventilation	TV 4–6 (≤ 8) mL/kg PBW, plateau ≤ 30 cmH_2O, adequate PEEP.
VAP bundle	HOB 30–45°, sedation vacation, chlorhexidine oral care, PUD & DVT prophylaxis.
DOPE	Displacement, Obstruction, Pneumothorax, Equipment failure.
Contraindication	Futile cases (no chance of recovery / DNR).

Invasive vs non-invasive	Invasive: ETT / tracheostomy. Non-invasive: face mask, helmet (CPAP/BiPAP); HFNC gives oxygen.
A/C	Full support: set volume for every breath (patient- or machine-triggered).
PSV	Pressure boost for each spontaneous breath; patient sets rate.
CMV	Machine controls all breaths; no spontaneous breathing.
Complications	Barotrauma/pneumothorax, volutrauma, auto-PEEP, oxygen toxicity, VILI, VAP, hypotension, stress ulcer, DVT.
Alarms	High pressure: secretions, biting, kink, bronchospasm. Low pressure: disconnection, cuff leak.
Cuff pressure	20–30 cmH_2O.

LECTURE 4

Pulmonary Edema

الارتشاح الرئوي

Part A · Past Exam Questions

أسئلة الدفعات السابقة

4.1 Pulmonary edema is characterized by which of the following? 1. Frothy secretions 2. Induced by heart failure 3. Induced by bronchial asthma 4. Tachypnea

كويزات سابقة (طب حالات حرجة)

a. 1,3,4
b. 1,2,4
c. 2,3,4

4.2 Which one of the following is a management option for pulmonary edema?

كويزات سابقة (طب حالات حرجة)

a. Performing a thoracoscopy
b. Administering specific antibiotic
c. Putting the patient on CPAP mode
d. Administering anticoagulant drugs

4.3 What is the most common cause of cardiogenic pulmonary edema?

أسئلة سابقة

a. Pneumonia
b. Acute pancreatitis
c. Left ventricular failure
d. Sepsis

4.4 Which of the following is NOT a direct lung injury that can lead to non-cardiogenic pulmonary edema?

أسئلة سابقة

a. Pneumonia
b. Sepsis
c. Inhalational injury
d. High altitude pulmonary edema

4.5 Which of the following is characteristic of pulmonary edema?

أسئلة سابقة

a. Sudden chest pain and loss of consciousness
b. Frothy sputum and shortness of breath
c. Pain in the lower extremities
d. Decreased lung sounds on auscultation

4.6 Which diagnostic test is used to assess hypoxemia and hypercapnia in pulmonary edema?

أسئلة سابقة

a. Chest X-ray
b. Arterial Blood Gas (ABG)
c. Echocardiogram
d. CT Chest

4.7 What would you expect to see on a Chest X-ray (CXR) in a patient with pulmonary edema?

أسئلة سابقة

a. Hyperinflated lungs with flat diaphragm
b. Increase in pulmonary parenchymal opacification
c. Ground glass opacification
d. Normal lung fields

أسئلة سابقة

4.8 **Which of the following is most commonly associated with non-cardiogenic pulmonary edema?**

a. Myocardial infarction
b. Pneumonia
c. Hypertension
d. Congestive heart failure

أسئلة سابقة

4.9 **Which is a typical symptom of pulmonary edema?**

a. Increased sputum production
b. Severe chest pain
c. Cyanosis and sudden collapse
d. Shortness of breath (dyspnea (

أسئلة سابقة

4.10 **Which of the following is a characteristic finding in the physical examination of pulmonary edema?**

a. Bilateral wheezes without crepitations
b. Congested neck veins
c. Decreased respiratory rate
d. Unilateral reduced breath sounds

أسئلة سابقة

4.11 **What is the role of BNP (B-type natriuretic peptide) in diagnosing pulmonary edema?**

a. Detects myocardial infarction
b. Indicates the stretch of the left atrium in case of heart failure
c. Detects bacterial infection
d. Measures oxygen levels in the blood

أسئلة سابقة

4.12 **In the context of pulmonary edema, what is the purpose of using a Pulse Oximeter?**

a. To evaluate heart rhythm
b. To assess lung compliance
c. To measure oxygen saturation in the blood
d. To check for fluid buildup in the lungs

أسئلة سابقة

4.13 **How is cardiogenic pulmonary edema differentiated from non-cardiogenic pulmonary edema?**

a. By using a chest X-ray
b. By using an echocardiogram
c. By measuring arterial blood gases
d. By performing a bronchoscopy

أسئلة سابقة

4.14 **What is the most common cause of non-cardiogenic pulmonary edema?**

a. Pneumonia
b. Sepsis
c. Hypertensive crisis
d. Myocardial infarction

أسئلة سابقة

4.15 **What medication is used to treat pulmonary edema by reducing fluid overload?**

a. Diuretics
b. Antibiotics
c. Steroids
d. Bronchodilators

أسئلة سابقة

4.16 **Which treatment is used for severe pulmonary edema cases that require ventilatory support?**

a. Antihypertensive medication
b. Mechanical ventilation
c. Antibiotic therapy
d. Bronchodilator therapy

أسئلة سابقة

4.17 What does the term "frothy secretions" in pulmonary edema refer to?

a. Blood-stained sputum
b. Mucous secretion mixed with air
c. Thick yellow sputum
d. Normal clear sputum

أسئلة سابقة

4.18 In the case of unstable hemodynamics due to pulmonary edema, which treatment is indicated?

a. Diuretics
b. Defibrillation
c. Mechanical ventilation
d. Oxygen therapy

أسئلة سابقة

4.19 Which of the following findings on CT chest is characteristic of pulmonary edema?

a. Hyperinflated lungs
b. Ground glass opacification
c. Consolidation
d. Pneumothorax

أسئلة سابقة

4.20 What is the key difference between cardiogenic and non-cardiogenic pulmonary edema?

a. Cardiogenic edema responds to diuretics, whereas non-cardiogenic usually does not
b. Cardiogenic edema is caused by pneumonia, while non-cardiogenic is due to heart failure
c. Non-cardiogenic edema is treated with corticosteroids, cardiogenic is not
d. Non-cardiogenic edema typically requires mechanical ventilation, cardiogenic does not

أسئلة سابقة

4.21 Which of the following conditions is NOT a cause of non-cardiogenic pulmonary edema?

a. Sepsis
b. High altitude pulmonary edema
c. Hypertension
d. Pneumonia

أسئلة سابقة

4.22 Which treatment is generally NOT used in the management of pulmonary edema?

a. Oxygen therapy
b. Diuretics
c. Fluid overload management
d. Steroids

أسئلة سابقة

4.23 What is a common sign found on a CXR for patients with pulmonary edema?

a. Increased lung parenchymal opacification
b. Hyperinflated lungs
c. Ground glass opacification
d. Pneumothorax

أسئلة سابقة

4.24 In which case is mechanical ventilation most likely required in pulmonary edema?

a. Mild shortness of breath
b. Severe dyspnea with hypoxemia
c. Cough with little sputum production
d. No hypoxemia

أسئلة سابقة

4.25 Which of the following is NOT a recommended preventive measure for pulmonary edema?

a. Regular check-ups with a cardiologist
b. Maintaining a healthy weight
c. Avoiding smoking
d. Consuming a high-salt diet

4.26 Which of the following is a key diagnostic tool for differentiating between cardiogenic and non- cardiogenic pulmonary edema? أسئلة سابقة

a. CT Chest
b. Echocardiogram
c. Arterial Blood Gas
d. Chest X-ray

4.27 What is the role of diuretics in the treatment of pulmonary edema? أسئلة سابقة

a. To reduce inflammation in the lungs
b. To increase oxygen saturation in the blood
c. To reduce fluid overload and relieve pressure on the heart
d. To decrease bronchospasm

4.28 Which of the following is a typical finding in a patient with cardiogenic pulmonary edema? أسئلة سابقة

a. Normal lung compliance
b. Increased lung interstitial fluid
c. Ground glass opacification
d. Unilateral chest pain

4.29 Which of the following conditions can lead to non-cardiogenic pulmonary edema? أسئلة سابقة

a. Myocardial infarction
b. Aspiration of gastric contents
c. Severe arrhythmias
d. Hypertension

4.30 What is the most appropriate action for a patient with pulmonary edema caused by sepsis? أسئلة سابقة

a. Diuretics to relieve fluid buildup
b. Treat the underlying infection with antibiotics
c. Administer mechanical ventilation
d. Administer corticosteroids

4.31 Which of the following is true about pulmonary edema treatment? أسئلة سابقة

a. Fluids are used to treat pulmonary edema
b. Oxygen therapy is indicated for all patients
c. Diuretics are contraindicated in non-cardiogenic pulmonary edema
d. Bronchodilators are the first-line treatment

4.32 In a patient with pulmonary edema and severe hypoxemia, what is the target SpO2 range? أسئلة سابقة

a. 80-85%
b. 88-92%
c. 95-100%
d. 75-80%

4.33 What is a key difference between ARDS and cardiogenic pulmonary edema? أسئلة سابقة

a. ARDS is primarily caused by heart failure, while cardiogenic edema is caused by infections
b. ARDS is characterized by bilateral infiltrates on CXR, which is not seen in cardiogenic edema
c. ARDS patients do not require intubation, while cardiogenic edema requires mechanical ventilation
d. ARDS responds to high-volume fluid resuscitation, unlike cardiogenic pulmonary edema

4.34 Which of the following findings would support a diagnosis of ARDS rather than cardiogenic pulmonary edema? أسئلة سابقة

a. Presence of pleural effusion
b. Presence of bilateral infiltrates on CXR
c. Increased oxygenation despite high PEEP
d. History of recent myocardial infarction

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

4.N1 The main mechanism of cardiogenic pulmonary edema is: NCLEX

a. Increased pulmonary capillary hydrostatic pressure from left ventricular failure
b. Increased capillary permeability from sepsis
c. Low oncotic pressure only
d. Airway obstruction

4.N2 Classic signs of acute pulmonary edema include (select all): SELECT ALL

a. Severe dyspnea and orthopnea
b. Pink frothy sputum
c. Bilateral fine crackles
d. Anxiety and diaphoresis
e. Bradypnea

4.N3 The best position for a patient with acute pulmonary edema is: NCLEX

a. Supine with legs elevated
b. High Fowler's with legs dependent
c. Prone
d. Trendelenburg

4.N4 Treatment of acute cardiogenic pulmonary edema includes (select all): SELECT ALL

a. Oxygen / CPAP or BiPAP
b. IV furosemide
c. Nitrates if BP allows
d. Rapid IV fluid boluses
e. Morphine (cautiously) in selected cases

4.N5 After IV furosemide, the nurse should monitor closely for: NCLEX

a. Hypokalemia and hypotension
b. Hyperkalemia
c. Hyperglycemia only
d. Weight gain

4.N6 A BNP level that is markedly elevated in a dyspneic patient suggests: NCLEX

a. Cardiac (heart failure) cause
b. Pneumonia
c. Asthma
d. Pulmonary embolism only

4.N7 Chest X-ray findings of cardiogenic pulmonary edema include (select all): SELECT ALL

a. Cardiomegaly
b. Kerley B lines
c. Bat-wing perihilar opacities
d. Pleural effusions
e. Hyperinflated lungs

4.N8 **Non-invasive CPAP helps in pulmonary edema by (select all):** SELECT ALL

a. Recruiting alveoli and improving oxygenation
b. Reducing preload
c. Reducing LV afterload
d. Increasing venous return
e. Reducing work of breathing

4.N9 **Causes of non-cardiogenic pulmonary edema include (select all):** SELECT ALL

a. Sepsis
b. Aspiration
c. High altitude
d. Near-drowning
e. Mitral stenosis

4.N10 **Nitroglycerin is withheld in pulmonary edema if:** NCLEX

a. Systolic BP < 90 mmHg
b. Heart rate is 90
c. SpO_2 is 92%
d. Patient is anxious

4.N11 **Indications for intubation in acute pulmonary edema include (select all):** SELECT ALL

a. Failure of NIV
b. Decreased level of consciousness
c. Severe respiratory acidosis or exhaustion
d. Hemodynamic instability
e. Mild dyspnea on exertion

4.N12 **Teaching to prevent recurrence of cardiogenic pulmonary edema includes (select all):** SELECT ALL

a. Low-salt diet and fluid restriction as ordered
b. Daily weight; report gain > 1–2 kg in a day
c. Take medications regularly
d. Stop diuretics when feeling well
e. Report increasing dyspnea or orthopnea

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

Pulmonary edema	Fluid accumulation in the lung interstitium and alveoli → impaired gas exchange.	**Cardiogenic**	LV failure, MI, valve disease, hypertensive crisis → raised hydrostatic pressure.
Non-cardiogenic	Increased permeability: sepsis (most common), pneumonia, aspiration, inhalation injury, high altitude.	**Symptoms**	Severe dyspnea, orthopnea, PND, frothy (pink) sputum, cough, anxiety.
Signs	Tachypnea, tachycardia, bilateral crackles ± wheeze, congested neck veins, cyanosis.	**Investigations**	ABG, pulse oximetry, CXR (↑ parenchymal opacity), CT (ground glass), ECG, BNP, echo.

Echo	Key test to differentiate cardiogenic from non-cardiogenic.
Management	Sit up, O_2 (SpO_2 88–92% target in severe hypoxemia), CPAP/NIV, diuretics, nitrates, treat cause; MV if failing.
Sepsis-induced	Treat infection with antibiotics; diuretics less useful.
ARDS vs cardiogenic	ARDS: bilateral infiltrates, normal heart size, no response to diuretics, refractory hypoxemia.

BNP	Released with ventricular / atrial stretch — high in cardiogenic.
Diuretics	Furosemide reduces fluid overload; watch K^+ and BP.
Not used	Steroids are not routine treatment; avoid high-salt diet.

Answer Key — All Lectures

الإجابات

1.1. b — False
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

1.2. c — Coronary angiography and coronary revascularization
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

1.3. d — Aminophylline
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

1.4. c — Checking cardiac enzymes
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

1.5. a — Coronary angiography & revascularization
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

1.6. b — False
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

1.7. c — Sodium concentration
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

1.8. c — Cardiac enzymes
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

1.9. b — A spectrum of conditions including unstable angina, NSTEMI, and STEMI

1.10. b — Chest pain, often described as pressure, tightness, or heaviness

1.11. b — Cardiac Enzymes (e.g., Troponin)

1.12. b — T-wave inversion or ST segment depression

1.13. b — NSTEMI involves partial occlusion of a coronary artery, whereas STEMI involves complete occlusion

1.14. a — Physical activity, emotional stress, or cold weather

1.15. b — Nitroglycerin

1.16. b — The prognosis of death and ischemic events in ACS patients

1.17. b — The risk of in-hospital mortality

1.18. b — Severe chest pain with radiation, nausea, and sweating

1.19. a — Significant ST-segment elevation in ≥2 contiguous leads

1.20. b — Myocardial necrosis

1.21. a — The pain occurs at rest or with minimal exertion in UA

1.22. b — MONA (Morphine, Oxygen, Nitroglycerin, Antithrombotic therapy)

1.23. b — To reduce myocardial oxygen demand

1.24. a — Coronary Angiography followed by PCI (Percutaneous Coronary Intervention)

1.25. a — Coronary Artery Bypass Grafting (CABG)

1.26. a — V-A ECMO
كمان بيتستخدم كدعم IABP خلي بالك إن الـ V-A ECMO. الملف اختار ميكانيكي في الـ ACS.

1.27. a — Impella

1.28. b — Improves blood flow to the heart and reduces workload during PCI

1.29. c — ST-segment depression or T-wave inversion

1.30. a — It is a classification of heart failure severity, which impacts mortality risk

1.31. b — Risk of in-hospital mortality

1.32. b — Thrombolytics

1.33. b — Atherosclerotic plaque rupture and thrombus formation

1.34. b — It provides pain and anxiety relief

1.35. b — Prevent thrombus formation by inhibiting platelet aggregation

1.36. b — Elevated troponin levels

1.37. b — Transient reversible ischemia

1.38. b — Low blood pressure with signs of organ dysfunction

1.39. c — Proceed with coronary angiography and revascularization

1.N1. c — NSTEMI
ST depression/T inversion + raised troponin = NSTEMI. Same ECG with normal troponin = unstable angina.

1.N2. a, b, c — Unstable angina; NSTEMI; STEMI
Stable angina is chronic coronary disease, not ACS.

1.N3. b — Troponin I/T
Troponin rises 3–4 hrs, peaks 24 hrs, stays up 7–14 days.

1.N4. a — ≤ 90 minutes
If PCI is not available, give fibrinolytics with door-to-needle ≤ 30 minutes.

1.N5. a, b, c — Systolic BP > 90 mmHg; No recent phosphodiesterase inhibitor (sildenafil); No right ventricular infarction
Nitrates can cause severe hypotension in these situations.

1.N6. a — Aspirin 162–325 mg
Chewing speeds absorption; dual antiplatelet therapy adds clopidogrel/ticagrelor.

1.N7. a, b, c, e — Previous hemorrhagic stroke; Active internal bleeding; Suspected aortic dissection; Recent major surgery or trauma
Age alone is not a contraindication.

1.N8. b — Reducing myocardial oxygen demand
Avoid in heart failure, bradycardia, hypotension or cardiogenic shock.

1.N9. d — Cardiogenic shock
Class I none, II crackles/S3, III pulmonary edema, IV cardiogenic shock.

1.N10. a, b, c — Relief of chest pain; Resolution of ST elevation; Reperfusion arrhythmias
Accelerated idioventricular rhythm is a typical reperfusion arrhythmia.

1.N11. b — Continuous ECG monitoring with IV access and pain relief
Life-threatening arrhythmias (VF) are most common early.

1.N12. a — Atypical symptoms such as dyspnea, fatigue or epigastric pain
Silent or atypical MI is common in these groups.

2.1. a — True
إجابة مقترحة (المصدر مفيهوش إجابة رسمية). الطالب في الصورة هما الـ VF والـ VT pulseless الـ True: بس الصح ،False اختار shockable rhythms. أما الـ VT اللي معاها نبض فبتتعالج بـ cardioversion مش defibrillation.

2.2. a — True
إجابة مقترحة (المصدر مفيهوش إجابة رسمية). الطالب اختار False، تشخيصه أصلاً بالـ STEMI الـ True: بس الصح ECG (ST elevation في ليدين متجاورين أو أكتر).

2.3. b — Ventricular Fibrillation
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

2.4. b — Pulseless ventricular tachycardia
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

2.5. b — False
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

2.6. b — Sinus tachycardia

2.7. c — Electrocardiogram (ECG)

2.8. c — Ventricular Fibrillation (VF)

2.9. a — Shockable rhythms require defibrillation, non-shockable rhythms do not

2.10. c — Perform defibrillation

2.11. c — Never

2.N1. a — 0.04 seconds
Large square = 0.2 s (5 small squares); 5 large squares = 1 s.

2.N2. b — Atrial depolarization
QRS = ventricular depolarization; T wave = ventricular repolarization.

2.N3. b — 0.12–0.20 s
PR > 0.20 s = first-degree AV block.

2.N4. a — < 0.12 seconds
A wide QRS (≥ 0.12 s) suggests a ventricular origin or bundle branch block.

2.N5. b — 100 bpm
1500 ÷ 15 = 100. With large squares: 300 ÷ number of large squares.

2.N6. c — Count QRS complexes in a 6-second strip × 10
The 6-second method works for irregular rhythms like AF.

2.N7. a — 4th intercostal space, right sternal border
V2 = 4th ICS left sternal border; V4 = 5th ICS midclavicular.

2.N8. a, b, c, d — Determine the rate; Assess regularity; Examine P waves; Measure PR and QRS
Always assess the patient too: is there a pulse?

2.N9. a — < 60 bpm
Treat if symptomatic: atropine 1 mg IV first-line.

2.N10. b — Atrial fibrillation
AF raises the risk of thrombo-embolic stroke → anticoagulation.

2.N11. a — Atrial flutter
Atrial rate about 300/min.

2.N12. b — Premature ventricular contractions (PVCs)
Frequent or multifocal PVCs, couplets or R-on-T can lead to VT/VF.

2.N13. b — Check the patient's pulse and responsiveness
Pulseless VT → defibrillate; VT with pulse & unstable → synchronized cardioversion; stable → amiodarone.

2.N14. a — Ventricular fibrillation
VF = no cardiac output → immediate defibrillation and CPR.

2.N15. b — Check the patient and make sure everyone is clear
Also check lead/monitor artefact: a loose lead can mimic VF or asystole.

2.N16. b — Check the patient, leads and gain; confirm in another lead
Asystole is non-shockable: CPR + epinephrine 1 mg every 3–5 min.

2.N17. a — Organized electrical activity on the monitor without a palpable pulse
Search for reversible causes (Hs and Ts).

2.N18. a, b — Ventricular fibrillation; Pulseless ventricular tachycardia
Asystole and PEA are treated with CPR and epinephrine.

2.N19. a — P waves and QRS complexes beating independently (AV dissociation)
Treatment: pacing (transcutaneous then transvenous).

2.N20. a — Mobitz I (Wenckebach) second-degree block
Mobitz II: constant PR with sudden dropped beats — more dangerous.

2.N21. b — Hyperkalemia
Hypokalemia gives flat T waves and U waves.

2.N22. a — ST elevation in ≥ 2 contiguous leads or new LBBB
ST elevation in II, III, aVF = inferior MI; V1–V4 = anterior MI.

2.N23. a, b, c, d — Clean and dry skin before electrode placement; Change electrodes every 24 hours or as needed; Set alarm limits and never switch them off; Check the patient, not just the monitor
Treat the patient, not the monitor.

2.N24. a — Delivers the shock on the R wave
Used for unstable tachycardias with a pulse (AF, flutter, SVT, VT with pulse).

2.N25. a — > 100 bpm
Treat the cause: pain, fever, hypovolemia, anxiety, hypoxia.

3.1. a — Invasive or non-invasive
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

3.2. a — True
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

3.3. b — 4-6
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

3.4. c — 2,3,4
إجابة مقترحة (المصدر مفيهوش إجابة رسمية). إجابة الطالب في صورة الكويز. المحاضرة القديمة بتعد الـ modes: A/C, SIMV, PSV, CPAP, CMV.

3.5. c — 30 to 45 degrees

3.6. c — To prevent respiratory and circulatory complications

3.7. a — Measures for preventing respiratory and circulatory complications in ventilated patients

3.8. a — To prevent oxygenation failure

3.9. b — To support lung function without intubation

3.10. b — Low tidal volumes and proper PEEP

3.11. b — Oxygenation and ventilation parameters

3.12. b — To help a patient breathe when they cannot breathe on their own

3.13. b — High Flow Nasal Cannula (HFNC)

3.14. b — Insufficient oxygenation (hypoxemia)

3.15. b — It aids in the removal of carbon dioxide (CO_2)

3.16. b — Patient at risk of aspiration

3.17. a — Injury from excessive tidal volume and airway pressure

3.18. a — Incomplete exhalation leading to positive pressure build-up

3.19. d — External Ventilation Mask

3.20. b — Respiratory failure due to airway constriction

3.21. b — To provide full ventilator support if the patient cannot breathe on their own

3.22. b — Synchronous Intermittent Mandatory Ventilation (SIMV)

3.23. c — A patient who can breathe independently but needs assistance with the effort

3.24. b — It keeps the airways open without controlling the patient's breaths

3.25. b — The ventilator provides all breaths with no spontaneous breathing allowed

3.26. d — Increased oxygen saturation

3.27. b — To prevent the development of stress ulcers

3.28. d — All of the above

3.29. a — High levels of oxygen delivery

3.30. c — A patient with severe airway obstruction or respiratory failure

3.31. c — Facilitates carbon dioxide removal

3.32. a — High PEEP and excessive tidal volume

3.33. c — Are starting to breathe independently but need some ventilator assistance

3.34. d — Acute respiratory distress syndrome (ARDS)
من أهم دواعي الـ ARDS وده غلط لأن الـ MV. الملف كاتب ARDS، الإجابة الصح c (cardiac arrest with no chance of recovery / DNR).

3.35. a — Assist/Control (A/C)

3.36. b — Oxygen toxicity

3.37. b — To assess whether sedation can be reduced or stopped

3.38. b — Pneumothorax

3.39. b — High Frequency alarm

3.40. d — All of the above

3.N1. a, b, c, d — Hypoxemic respiratory failure; Hypercapnic respiratory failure; Airway protection (GCS ≤ 8); Cardiac arrest
Inadequate oxygenation, ventilation or airway protection.

3.N2. a, b, c, e — Secretions in the airway; Patient biting the tube or coughing; Kinked tubing; Bronchospasm
Disconnection or cuff leak triggers the LOW-pressure alarm.

3.N3. a — Displacement, Obstruction, Pneumothorax, Equipment failure
If unsure, disconnect and bag with 100% O_2.

3.N4. a, b, c, d — Head of bed 30–45°; Daily sedation interruption and readiness-to-wean assessment; Oral care with chlorhexidine; DVT and peptic ulcer prophylaxis
Circuits are changed only when visibly soiled.

3.N5. a — 20–30 cmH_2O
Low → aspiration/leak; high → tracheal ischemia.

3.N6. a, b, c, e — Pre-oxygenate with 100% O_2; Apply suction only while withdrawing; Limit each pass to 10–15 seconds; Monitor SpO_2 and heart rate
Use sterile technique (or a closed in-line system).

3.N7. a — Keeping alveoli open at end-expiration
High PEEP can reduce venous return (hypotension) and cause barotrauma.

3.N8. a — Assist/Control (A/C)
Risk: hyperventilation if the patient breathes fast.

3.N9. a, b, c, d — Underlying cause improving; Hemodynamic stability; $FiO_2 \leq 40–50\%$ with PEEP ≤ 5–8; Adequate cough and mental status
Spontaneous breathing trial (T-piece or low PSV) for 30–120 minutes.

3.N10. a, b, c, d — RR > 35/min; $SpO_2 < 90\%$; Heart rate rising > 20% or arrhythmia; Diaphoresis and agitation
Stop the trial and return to previous settings.

3.N11. a, b, c, e — Waveform capnography ($ETCO_2$); Bilateral equal breath sounds and chest rise; Chest X-ray; Absence of gastric sounds
Note the tube mark at the lips/teeth.

3.N12. a — 4–6 (up to 8) mL/kg predicted body weight
Plateau pressure ≤ 30 cmH_2O.

4.1. b — 1,2,4
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

4.2. c — Putting the patient on CPAP mode
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

4.3. c — Left ventricular failure

4.4. b — Sepsis

4.5. b — Frothy sputum and shortness of breath

4.6. b — Arterial Blood Gas (ABG)

4.7. b — Increase in pulmonary parenchymal opacification

4.8. b — Pneumonia

4.9. d — Shortness of breath (dyspnea (

4.10. b — Congested neck veins

4.11. b — Indicates the stretch of the left atrium in case of heart failure

4.12. c — To measure oxygen saturation in the blood

4.13. b — By using an echocardiogram

4.14. b — Sepsis

4.15. a — Diuretics

4.16. b — Mechanical ventilation

4.17. b — Mucous secretion mixed with air

4.18. b — Defibrillation
بيكون لو DC shock (cardioversion) الإجابة دي من الملف زي ما هي. الـ arrhythmia عدم الاستقرار سببه

4.19. b — Ground glass opacification

4.20. a — Cardiogenic edema responds to diuretics, whereas non-cardiogenic usually does not

4.21. c — Hypertension

4.22. d — Steroids

4.23. a — Increased lung parenchymal opacification

4.24. b — Severe dyspnea with hypoxemia

4.25. d — Consuming a high-salt diet

4.26. b — Echocardiogram

4.27. c — To reduce fluid overload and relieve pressure on the heart

4.28. b — Increased lung interstitial fluid

4.29. b — Aspiration of gastric contents

4.30. b — Treat the underlying infection with antibiotics

4.31. b — Oxygen therapy is indicated for all patients

4.32. b — 88-92%

4.33. b — ARDS is characterized by bilateral infiltrates on CXR, which is not seen in cardiogenic edema

4.34. b — Presence of bilateral infiltrates on CXR

4.N1. a — Increased pulmonary capillary hydrostatic pressure from left ventricular failure
Non-cardiogenic (ARDS) = increased permeability.

4.N2. a, b, c, d — Severe dyspnea and orthopnea; Pink frothy sputum; Bilateral fine crackles; Anxiety and diaphoresis
Tachypnea, tachycardia and hypoxemia.

4.N3. b — High Fowler's with legs dependent
Reduces venous return and eases breathing.

4.N4. a, b, c, e — Oxygen / CPAP or BiPAP; IV furosemide; Nitrates if BP allows; Morphine (cautiously) in selected cases
Fluids worsen the overload.

4.N5. a — Hypokalemia and hypotension
Also strict intake/output and daily weight.

4.N6. a — Cardiac (heart failure) cause
BNP is released when ventricles are stretched.

4.N7. a, b, c, d — Cardiomegaly; Kerley B lines; Bat-wing perihilar opacities; Pleural effusions
Hyperinflation is typical of COPD/asthma.

4.N8. a, b, c, e — Recruiting alveoli and improving oxygenation; Reducing preload; Reducing LV afterload; Reducing work of breathing
Positive intrathoracic pressure reduces venous return.

4.N9. a, b, c, d — Sepsis; Aspiration; High altitude; Near-drowning
Mitral stenosis raises left atrial pressure → cardiogenic.

4.N10. a — Systolic BP < 90 mmHg
It reduces preload and afterload but causes hypotension.

4.N11. a, b, c, d — Failure of NIV; Decreased level of consciousness; Severe respiratory acidosis or exhaustion; Hemodynamic instability

4.N12. a, b, c, e — Low-salt diet and fluid restriction as ordered; Daily weight; report gain > 1–2 kg in a day; Take medications regularly; Report increasing dyspnea or orthopnea